Reader: The phrase "the values assumed on the set of students" sounds somewhat awkward.

Author: Because you are used to interpret "value" as "numerical value".

Let us consider some other examples.

Example 2: Let D be a set of all triangles, and E a set of positive real numbers. Using these sets, we can define two functions, namely, the area of a triangle and the perimeter of a triangle. Both functions are mappings (certainly, of different nature) of the set of the triangles onto the set of the positive real numbers. It is said that the set of all the triangles is the domain of these functions and the set of the positive real numbers is the range of these functions.

Scan: Q) $\forall x \in R$, does there exist a triangle having area $= x$ or perimeter $= x$?

Example 3: Let D be a set of all triangles, and E a set of all circles. The mapping of D onto E can be either a circle inscribed in a triangle, or a circle circumscribed around a triangle. **Domain:** Set of all triangles

Range: set of all circles.

By the way, do you think that it is possible to "construct" an inverse function in a similar way, namely, to define a function with all the circles as its domain and all the triangles as its range?

Reader: I see no objections.

Author: No, it is impossible. Because any number of different triangles can be inscribed in or circumscribed around a circle. ↳ Proof??

In other words, each element of E (each circle) corresponds to an infinite number of different elements of D (i.e. an infinite number of triangles). It means that there is no function since no mapping can be realised.

However, the situation can be improved if we restrict the set of triangles.

~~However, the situation can be improved if we restrict the set of triangles.~~

Reader: I guess I know how to do it. We must choose the set of all the equilateral triangles as the set D. Then it becomes possible to realise both a mapping of D onto E (onto the set of all the circles) and an inverse mapping, i.e. the mapping of E onto D,